Debian in your pocket

Android’s Linux Terminal, from the inside

by Seth House

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This is Debian

```
$ vulkaninfo --summary
    deviceName = PowerVR D-Series DXT-48-1536 MC1
    driverID   = DRIVER_ID_IMAGINATION_PROPRIETARY
```

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- crosvm — the same VMM as ChromeOS’s Crostini.

How it works

The terminal is a web page

```
WebView + xterm.js
  └─ vsock → socat → ttyd
       └─ setpriv --reuid=droid … /bin/bash -l
```

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`setpriv` is not a login program.

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- It surfaced two reboots later, as a graphics error.

It’s not VNC

One PCI ID

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$ cat /sys/class/drm/card0/device/uevent
PCI_ID=1AF4:1050
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virtio-gpu. Graphics API calls are forwarded to the phone’s GPU driver.

Demo: SuperTuxKart

```
OpenGL renderer: zink Vulkan 1.3(PowerVR D-Series …
    (IMAGINATION_PROPRIETARY))
```

One environment variable, 12x

vkmark, 1080p	**score**
Vendor default	26
`MESA_VK_WSI_DEBUG` unset	321

Demo: vkQuake

Vulkan-native. No zink.

When it breaks

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Isolation runs one way.

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```
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×6,338. The VM was a casualty, not the cause.

The VM can adb into its phone

```
adb pair    $gateway:<port> <code>
adb connect $gateway:<port>
```

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- You can raise the RAM, but you shouldn’t.
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- No hardware video decode.
- The LAN can’t reach the VM, but any Android app can.

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Then one Android update invalidated my notes — and my verification script.

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Verify, don’t assume — including the AI.